Multiple Yield-Curve Construction Using QuantLib

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Thesis Objectives & Agenda

1. Explore the evolution of yield-curve construction post-2008, with a focus on the Eurozone.
2. Show the bootstrapping process of €STR and EURIBOR curves using QuantLib.
3. Analyze implications of the multiple-curve framework for constructing Euribor forward curves.
4. Evaluate the impact of using different bootstrapping sets on curve stability.

LIBOR-OIS Spread from 2006 to 2009

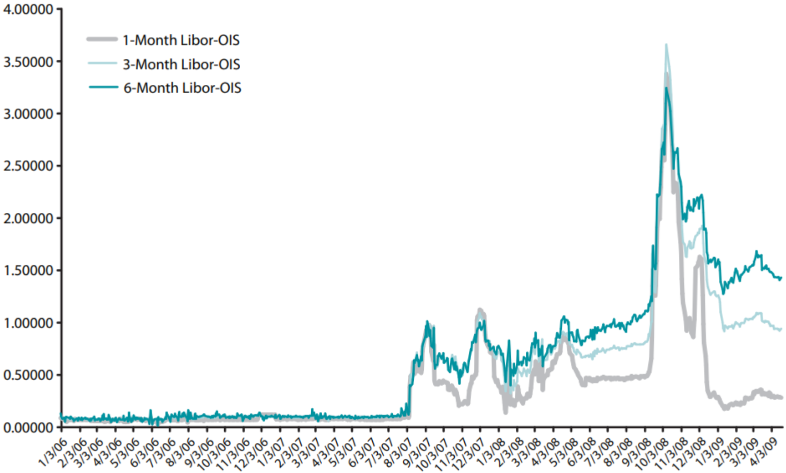

Figure: London Interbank Offered Rate (LIBOR) - OverNight Index Swap (OIS) spreads for different tenors during 2006-2009. Source: St. Louis (2009)

Post-Crisis Developments: A New Paradigm

- **Collateralization Impact:** increased use of collateral in swaps to manage counterparty risk influenced swap pricing, creating the need for an updated discounting method.
- **Post-2008 Response:** new bank lending programs established overnight rates as the standard for collateral remuneration.
- **New Risk-Free Proxy:** with overnight rates as the benchmark for collateral, OIS-based rates gradually became the risk-free standard for swap pricing.

The Multiple-Curve Framework

- **Dual-Curve Approach:** this approach uses two distinct curves:
  1. **Discounting Curve:** a risk-free zero curve, derived from OIS-based rates like the Euro Short Term Rate (€STR), used for discounting collateralized cash flows to present value.
  2. **Projection Curve:** a forward curve constructed from instruments tied to a specific European Interbank Offered Rate (EURIBOR) tenor, used to project future floating rates.
- **Tenor Consistency:** ensuring consistency between the tenors of market instruments and the projection curves is essential.

QuantLib and Methodology

- **QuantLib:** widely used in quantitative finance for pricing, modeling, and analysis. It was utilized for constructing the yield curves in this study.
- **Methodology:**
    1. **Instrument Selection:** instruments were selected to ensure consistency with yield curve maturities. Data was sourced as snapshot trading market quotes from Refinitiv Eikon on July 12, 2024.
    2. **Rate Helpers:** linked market quotes to curve points using QuantLib's RateHelper classes.
    3. **PiecewiseYieldCurve:** performed bootstrapping using PiecewiseLogCubicDiscount for smooth interpolation.

€STR Discount Curve Construction

Hierarchical bootstrapping with the following instruments:

1. **EUR Deposits**: short-term segment with ON, TN, SN.
2. **Short-term OISs**: short-term extension with one-week OIS.
3. **ECB Forward OISs**: incorporating expectations during ECB maintenance periods.
4. **Long-term OISs**: extending the curve up to 50 years.

ECB Forward OISs

- **Capturing Market Expectations:** included to reflect the market's forecast of ECB monetary policy decisions.
- **Hedging Tool:** used to manage interest rate risk during ECB reserve maintenance periods.

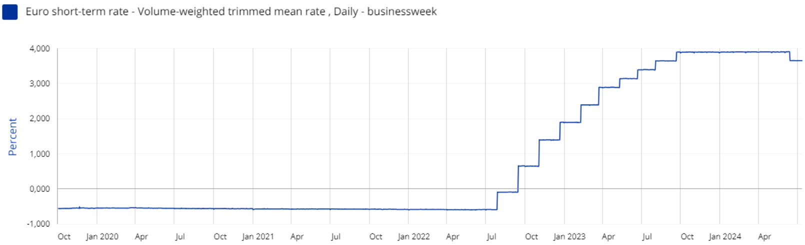

Figure: Historical series of €STR (2019-2024). Source: ECB (2024).

€STR Zero Curve

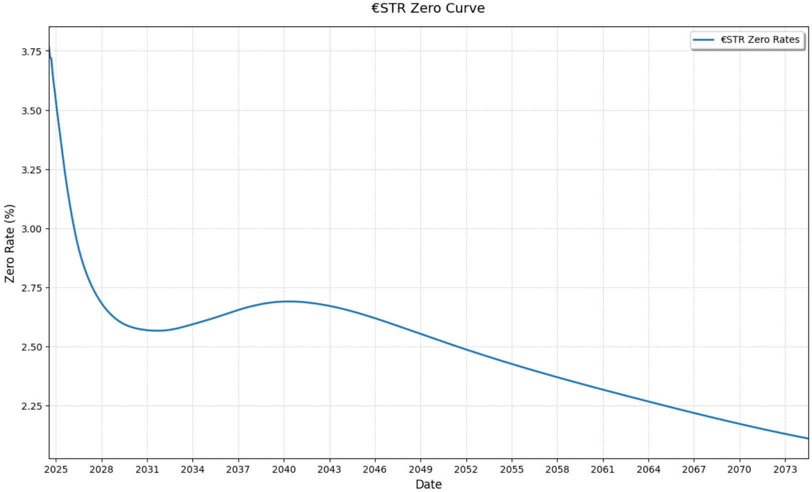

Figure: €STR zero curve (July 12, 2024-2074)

Liquidity Jumps: ECB Reserve Maintenance Periods and Turn-of-the-Year

- **ECB Reserve Maintenance Periods:** banks adjust liquidity towards the end of these periods to meet reserve requirements, possibly causing short-term rate spikes.
- **Turn-of-the-Year Effect:** anticipated year-end adjustments for regulatory reporting can lead to possible presence of short-term rate jumps.

Example: Turn-of-the-Year Jump Effect

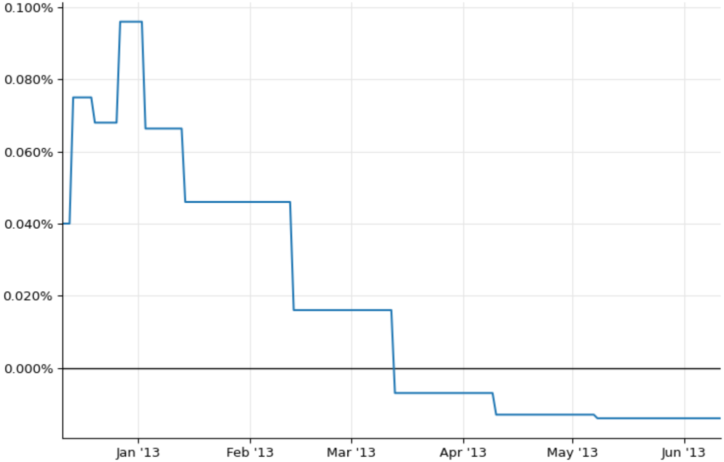

Figure: Turn-of-the-Year Jump in EONIA overnight forward rates (2012-2013). Source: Ametrano and Bianchetti (2013) and Ballabio and Balaraman (2023).

€STR Forward Rates and ECB Forward OISs

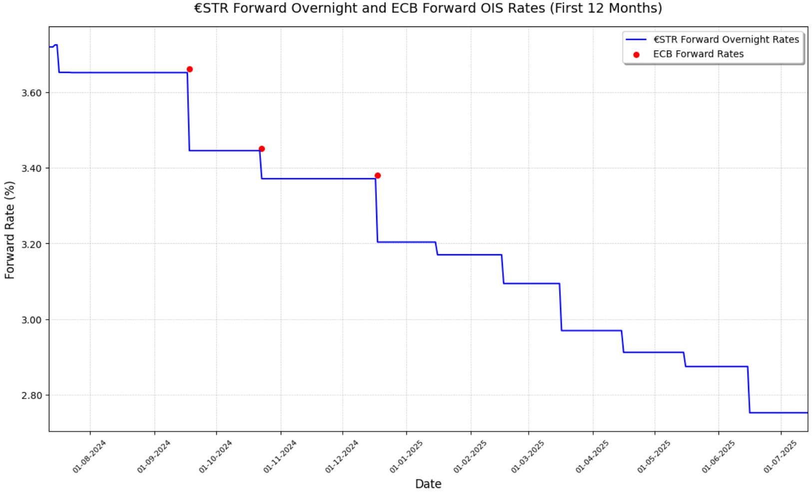

Figure: €STR overnight forward rates and ECB forward OISs for the first 12 months from July 12, 2024.

Euribor 3M Forward Curve Construction

Hierarchical bootstrapping with the following instruments:

1. **Synthetic Deposits**: derived from the Euribor-OIS/€STR basis, used to enhance accuracy in the very short-term segment.
2. **FRA**: 0X3 FRA used as the initial anchor point for the short-term segment (0-3 months).
3. **Futures**: Euribor 3M futures to extend the curve into the medium-term horizon (3 months - 2 years).
4. **Swaps**: Euribor 3M swaps to extend the curve into the long-term horizon (3-50 years).

Limitations and Role of Synthetic Deposits in Euribor 3M Curve Construction

- **Short-Term Data Gap:** the absence of Euribor 3M instruments with maturities shorter than 3 months leads to backward interpolation, which can misestimate forward rates in the initial curve segment.
- **Synthetic Deposits Solution:** constructed by modeling the Euribor-OIS/€STR basis, fill this gap by introducing additional short-term data points, enhancing curve accuracy in the initial segment.
- For a detailed derivation, see Appendix A.

Euribor 3M Forward Curve Without Synthetic Deposits

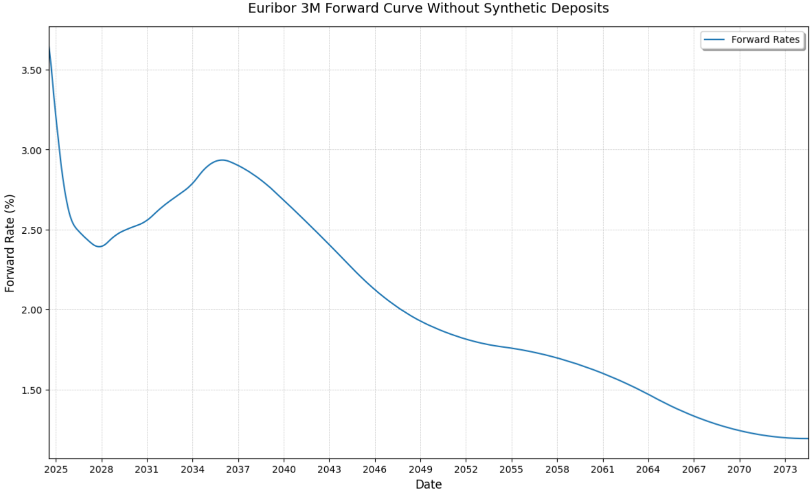

Figure: Euribor 3M forward curve constructed without synthetic deposits (July 12, 2024-2074).

Comparison With and Without Synthetic Deposits

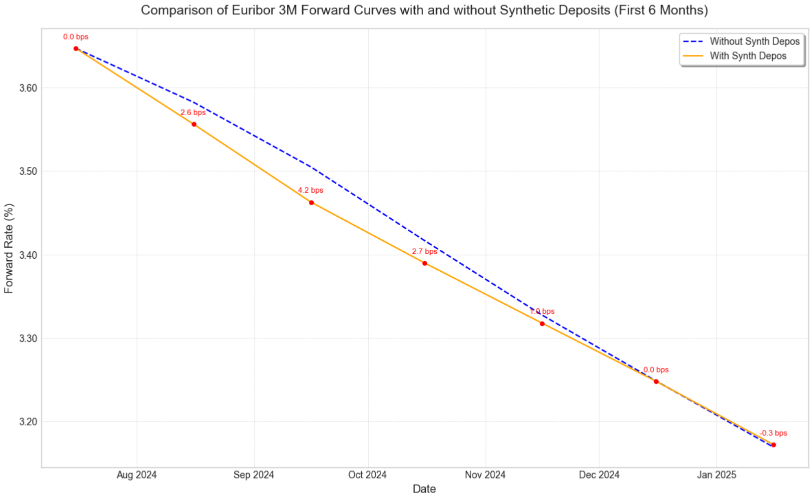

Figure: Comparison of Euribor 3M forward curves with and without synthetic deposits in the first 6 months from July 12, 2024.

Testing the Effects of the Multiple-Curve Framework

Objective 1: Test the Single vs. Dual Curve Approach

Show how the dual-curve approach impacts Euribor 3M forward curves compared to the single-curve approach which uses the same Euribor 3M for discounting.

Objective 2: Test the Tenor Consistency

Compare the Euribor 6M forward curve built with consistent instruments to one constructed using mismatched tenors (3M), highlighting the impact on curve accuracy.

For detailed instruments, see Appendix B

Test 1: Single vs Dual Curve Bootstrapping

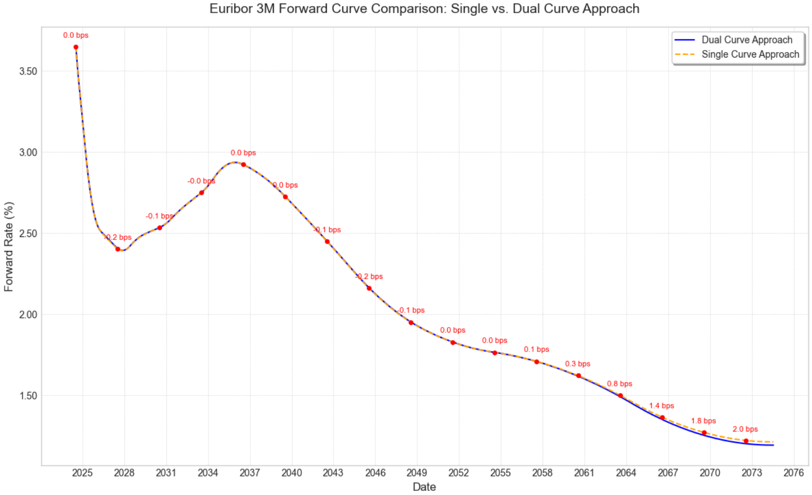

Figure: Comparison of Euribor 3M forward curves using single vs dual curve approaches (July 12, 2024-2074).

Test 2: The Importance of Tenor Consistency

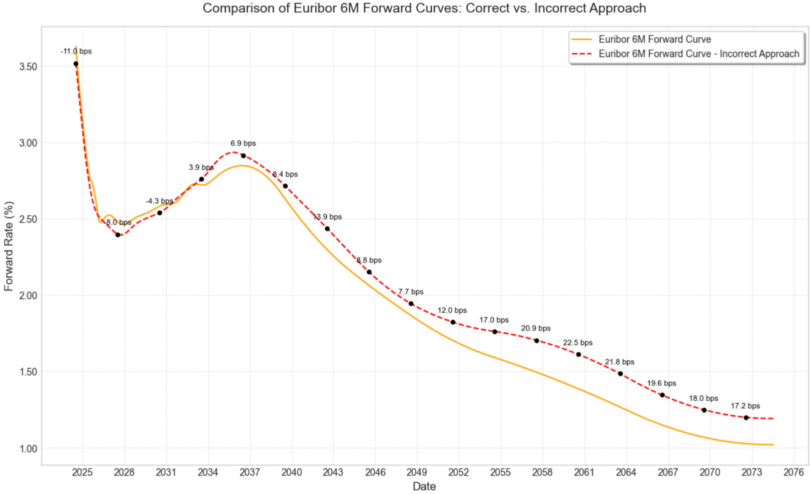

Figure: Euribor 6M forward curves - Comparison of the tenor consistent approach using 6M instruments vs incorrect approach using 3M instruments (July 12, 2024-2074).

Evaluating Different Bootstrapping Sets

- **The Choice of the Bootstrapping Set:** is highly influenced by the trader's expertise, as it requires balancing factors like liquidity, bid-ask spreads, and transaction costs. This decision-making process is context-dependent and relies on nuanced judgment.

It's more an art than a science!

The Four Bootstrapping Sets

Objective

Evaluate the impact of four different tenor-consistent bootstrapping sets on curve construction, with an additional focus on stability through bid/ask spread shock simulations.

- **Original Set:** synthetic deposits + 0X3 FRA + futures + swaps
- **Combination 2:** synthetic deposits + 0X3 FRA + futures + more focus on swaps
- **Combination 3:** synthetic deposits + 0X3 FRA + additional FRAs + swaps
- **Combination 4:** synthetic deposits + 0X3 FRA + additional FRAs + more focus on swaps

Comparative Analysis of Forward Curves

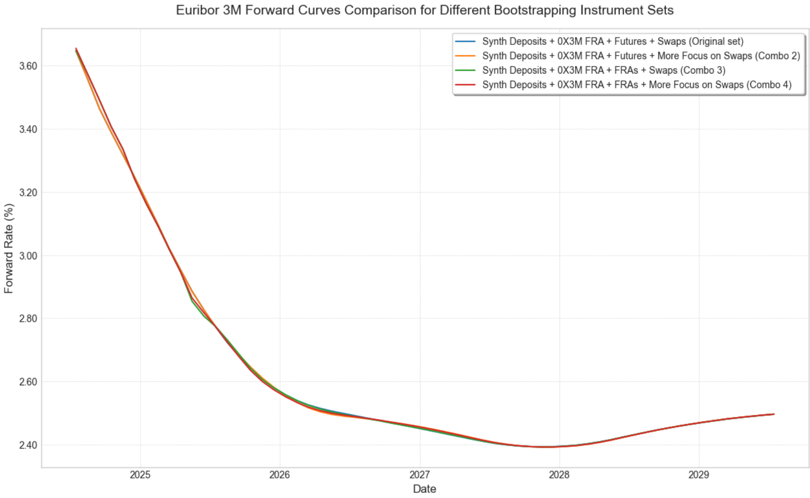

Figure: Comparison of four Euribor 3M forward curves builded with the different bootstrapping instrument sets (July 12, 2024-2029).

Stability Assessment: Bid/Ask Spread Shock Application

Objective

Assess the stability of the four forward curves by applying simulated bid/ask spread shocks to each instrument in the bootstrapping set.

- **Shock Simulation:** for each instrument, random shocks are generated based on its specific observed bid/ask spreads and applied to bid and ask prices.
- **Curve Reconstruction:** new adjusted mid-rates are recalculated for each shock scenario, with forward curves reconstructed in 1,000 simulations to evaluate stability.

For further details, see Appendix C.

Stability Assessment: Simulated Forward Curves

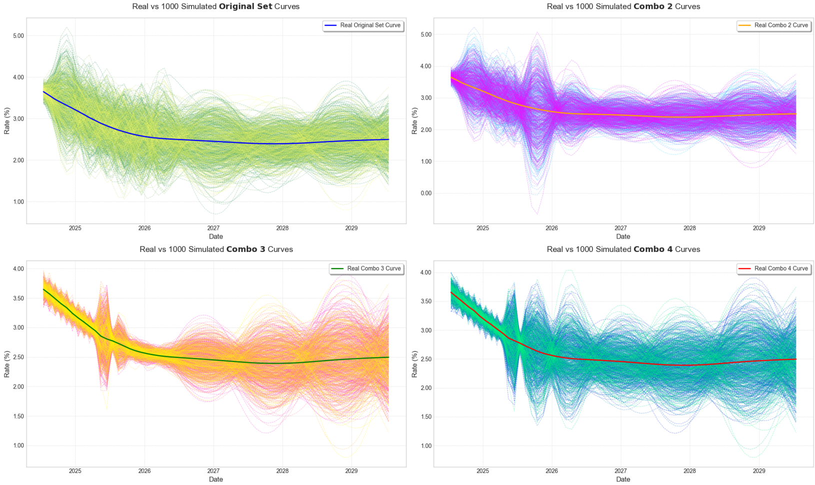

Figure: Comparison between the 'real' and one thousand shock-simulated Euribor 3M forward curves for each of the four bootstrapping sets (July 12, 2024-2029).

Stability Assessment: Conclusions

- **Higher Stability in Combo 3:** Combo 3 the lowest average Root Mean Square Deviation and lowest average max deviations.
- **FRAs vs. Futures:** FRA-focused sets (Combo 3 and 4) performed better, compared to futures-based sets (Original and Combo 2).

For detailed formulas, see Appendix D.

Bootstrapping Set	Avg RMSD (bps)	Avg Max-Positive Dev. (bps)	Avg Max-Negative Dev. (bps)	t-statistic	p-value
Combo 3	23.49	47.93	-47.80	73.76	0.0000
Combo 4	28.20	57.96	-57.29	86.30	0.0000
Combo 2	37.30	82.03	-80.56	88.69	0.0000
Original Set	38.38	77.40	-77.55	105.91	0.0000

Table: Stability metrics across the four bootstrapping sets under simulated market shocks.

Conclusions

- **Evolution of curve construction:** transition from single-curve to multiple-curve framework post-2008 crisis has transformed yield curve construction.
- **Instrument selection and bootstrapping:** tenor-consistent instrument sets are crucial for accurate curve construction, while incorrect sets can cause significant discrepancies.
- **Stability of yield curves:** bootstrapping sets focused on FRAs demonstrated higher resilience to bid/ask spread shocks.

Thank you for your attention!

References

Ametrano, Ferdinando M. and Marco Bianchetti (2013). "Everything You Always Wanted to Know About Multiple Interest Rate Curve Bootstrapping but Were Afraid to Ask". In: Available at SSRN: https://ssrn.com/abstract=2219548 or http://dx.doi.org/10.2139/ssrn.2219548.

Ballabio, Luigi and Goutham Balaraman (2023). *QuantLib Python Cookbook*. Leanpub. URL: https://leanpub.com/quantlibpythoncookbook.

St. Louis, Federal Reserve Bank of (2009). *What the Libor-OIS Spread Says*. Accessed: 2024-09-19. URL: https://files.stlouisfed.org/files/htdocs/publications/es/09/ES0924.pdf.

Appendix A: Synthetic Deposits Derivation (Part 1)

- The relationship between the Euribor forward rate $R_{\text{Eu}}(t_1, t_2)$ and the €STR forward rate $R_{\text{€STR}}(t_1, t_2)$ is given by:

$$R_{\text{Eu}}(t_1, t_2) \cdot \tau(t_1, t_2) = R_{\text{€STR}}(t_1, t_2) \cdot \tau(t_1, t_2) + \Delta(t_1, t_2)$$

  where $\Delta(t_1, t_2)$ is the cumulative basis over the period, defined as:

$$\Delta(t_1, t_2) = \int_{t_1}^{t_2} \delta(u)\, du$$

  and $\tau(t_1, t_2)$ is the year fraction for the period $[t_1, t_2]$.

- The instantaneous basis $\delta(t)$ can be approximated as a polynomial of degree n:

$$\delta(t) = \alpha + \beta \cdot t + \gamma \cdot t^2 + \dots$$

Appendix A: Synthetic Deposits Derivation (Part 2)

- If $\delta(t)$ is assumed to be constant ($n = 0$), the cumulative basis simplifies to:

$$\Delta(t_1, t_2) = \alpha \cdot \tau(t_1, t_2)$$

- Thus, the relationship becomes:

$$R_{\text{Eu}}(t_1, t_2) = R_{\text{€STR}}(t_1, t_2) + \alpha$$

- Here, α is the difference between Euribor and €STR rates:

$$\alpha = R_{\text{Eu}}(t_1, t_2) - R_{\text{€STR}}(t_1, t_2)$$

Once the basis is determined, we can add it to the €STR rates to obtain synthetic deposit rates $R(0, t)$ for any given maturity t.

Appendix B: Euribor 6M Forward Curve Construction

Correct Tenor-Consistent Approach: hierarchical bootstrapping using the following instruments:

1. **Synthetic Deposits (0-6m)**: constructed similarly to the 3-month curve using the Euribor-OIS/€STR basis, providing essential support for the initial segment of the curve.
2. **FRAs (6m-2y)**: starting with the 0X6 FRA as an anchor point, FRAs extend the curve into the medium-term horizon.
3. **Swaps (2y-50y)**: used to further extend the curve into the long-term horizon.

Incorrect Approach: the same market instruments as used for constructing the Euribor 3M forward curve are applied here, leading to tenor inconsistencies.

Appendix C: Application of the Bid/Ask Spread Shocks (Part 1)

- **Random Variable Generation:** for each instrument i, a random shock $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ was generated, with:

$$\sigma = \overline{S} = \frac{1}{N}\sum_{i=1}^{N} S_i$$

where $S_i = A_i - B_i$ (bid-ask spread), and N is the total number of instruments.

- **Shock Application:** shocks applied to bid and ask prices:

$$B_i^{\text{shocked}} = B_i(1+\varepsilon_i), \quad A_i^{\text{shocked}} = A_i(1+\varepsilon_i)$$

Ensuring a positive spread: adjust $A_i^{\text{shocked}} = B_i^{\text{shocked}} + |A_i^{\text{shocked}} - B_i^{\text{shocked}}|$ if necessary.

Appendix C: Application of the Bid/Ask Spread Shocks (Part 2)

- **Mid-Rate Calculation:** after applying shocks, calculate the new mid-rate:

$$M_i^{\text{shocked}} = \frac{B_i^{\text{shocked}} + A_i^{\text{shocked}}}{2}$$

- **Curve Reconstruction:** using M_i^{shocked}, we reconstructed the forward curves for each shock simulation. This process was repeated across 1,000 simulations to evaluate stability for each bootstrapping set.

Appendix D: Stability Metrics Formulas

- **Root Mean Square Deviation Calculation:** for each simulation k:

$$RMSD(k) = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(\hat{y}^t(k) - y_t)^2}$$

where T is the number of time points, $\hat{y}^t(k)$ is the simulated forward rate, and y_t is the real forward rate.

- **Max Deviations:** the average max-positive and max-negative deviations were also computed:

$$\text{Average Max-Positive Deviation} = \frac{1}{K}\sum_{k=1}^{K}\left(\max_t \left(\hat{y}^t(k) - y_t\right)\right)$$

$$\text{Average Max-Negative Deviation} = \frac{1}{K}\sum_{k=1}^{K}\left(\min_t \left(\hat{y}^t(k) - y_t\right)\right)$$